

ME416

Ground Vehicle Dynamics

Basic Vehicle Dynamics

[1]

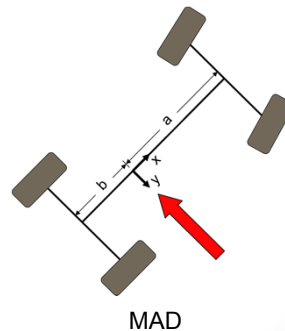
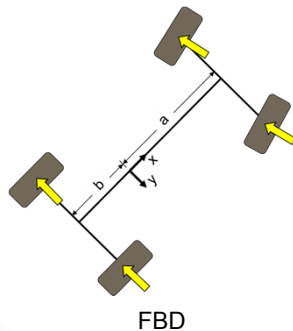
Introduction

- This is an extremely high level introduction of the topics we will cover in the first half of the course
- We will introduce many new words and concepts
- We will also be jumping around a bit and not fully developing concepts or definitions
- The concept is to give you an idea of what we need to consider when we design a ground vehicle
 - Allows you to pursue your own team project somewhat independently of the course lecture
 - Provides a frame of reference for subsequent more detailed lecture topics

[2]

Steady State Turning

- As a vehicle turns (steady state) the centripetal acceleration results from generation of lateral forces produced by the tires at the contact point with the ground
 - Are these lateral forces equally distributed at each corner?
 - How are these lateral forces generated?
 - How can we manipulate the response of the vehicle to our needs?



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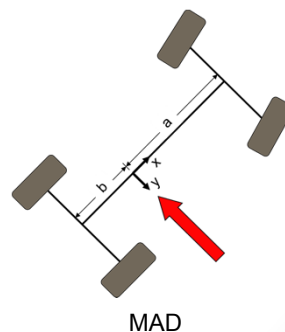
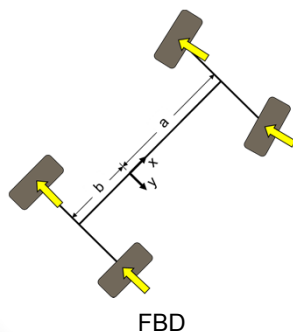
Steady State Turning (Cont.)

- Consider, the force balance for this simplified case (neutral)

$$F_{yFL} + F_{yFR} + F_{yRL} + F_{yRR} = m \cdot latG$$

and,

$$(F_{yFL} + F_{yFR}) \cdot a = (F_{yRL} + F_{yRR}) \cdot b$$



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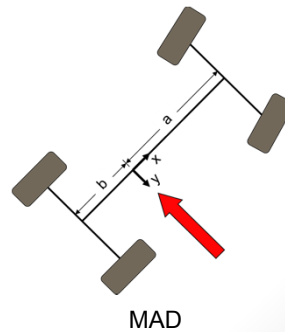
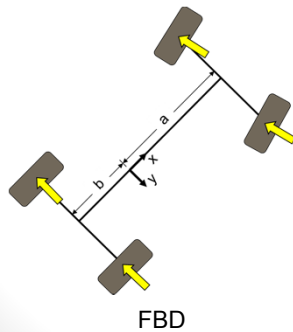
Steady State Turning (Cont.)

- Remember that statically on level ground the load on the front and rear will be

$$\frac{W_F}{W} = \frac{b}{WB} \quad \text{and} \quad \frac{W_R}{W} = \frac{a}{WB}$$

$$\frac{W_F}{W_R} = \frac{b}{a}$$

- Therefore, the normal load will not be equally distributed (statically) front to rear unless the center of gravity (c.g.) is at the midpoint between the front and rear axes

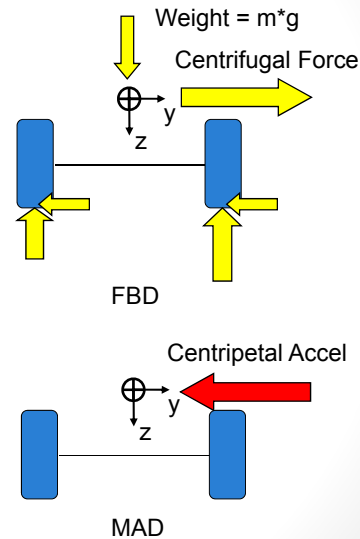


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Steady State Turning (Cont.)

- The c.g. of the vehicle is above the ground plane therefore there is a moment which will unload the inside tire and load the outside tire
- Therefore, the normal load on each tire will not be equal due to
 - The static weight distribution
 - The weight transfer due to the lateral acceleration
- But, the question was about the lateral forces generated by the tires
- To a first approximation we know that the lateral or friction force generated by the tire is directly proportional to the normal force



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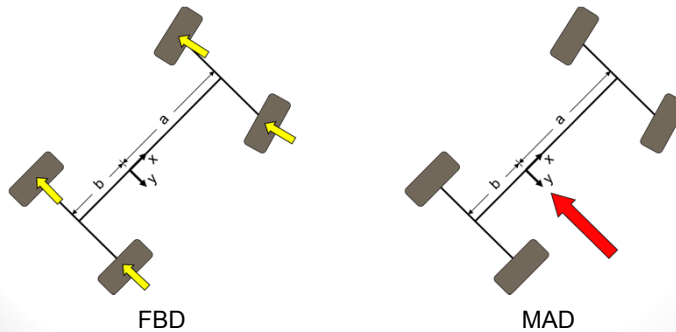
Steady State Turning (Cont.)

- So, now we have different normal loads at each corner and therefore different lateral forces generated, still with

$$F_{yFL} + F_{yFR} + F_{yRL} + F_{yRR} = m * latG$$

and,

$$(F_{yFL} + F_{yFR}) * a = (F_{yRL} + F_{yRR}) * b$$

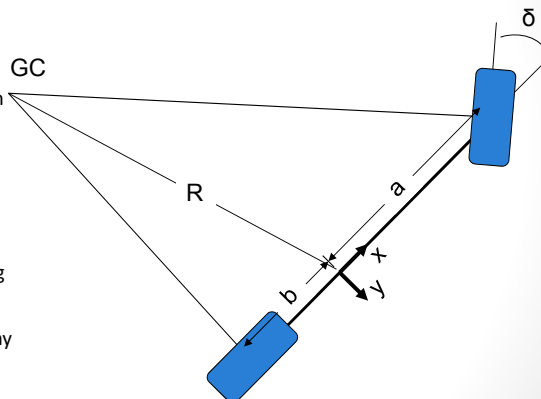


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Simple Two-Wheel (Bicycle) Model

- How are the lateral forces generated?
- Consider, very low speed turning, i.e. no lateral acceleration
 - Turning is entirely geometric and tires are rigid bodies
 - The wheels track the path with no deformation, i.e. zero slip
 - For the zero slip condition the geometric center (GC) of the turn must lie along the rolling axis of the unsteered rear wheel
 - The front wheel must be steered (δ) such that it's rolling axis passes through the geometric center of the turn
 - The tires are not generating any lateral forces (there are no lateral accelerations to counteract) and so are rolling only
 - Applies to very low speed situations such as parking lots etc.

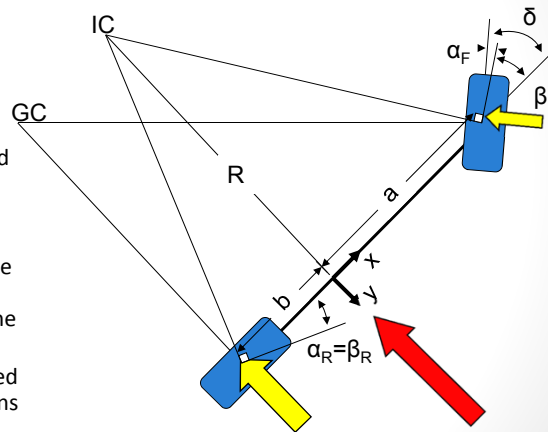


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Simple Bicycle Model – High Speed Turning

- For high speed turning, i.e. lateral acceleration, there must be lateral forces generated by the tires
 - The tires are no longer tracking about the geometric center (GC) and R is relative to the instantaneous center (IC)
 - The angle between the direction of heading of the tire (steer angle: δ) and direction of travel (β) is the slip angle (α)
 - Lateral forces are produced due to lateral deformations in the tire as it rolls

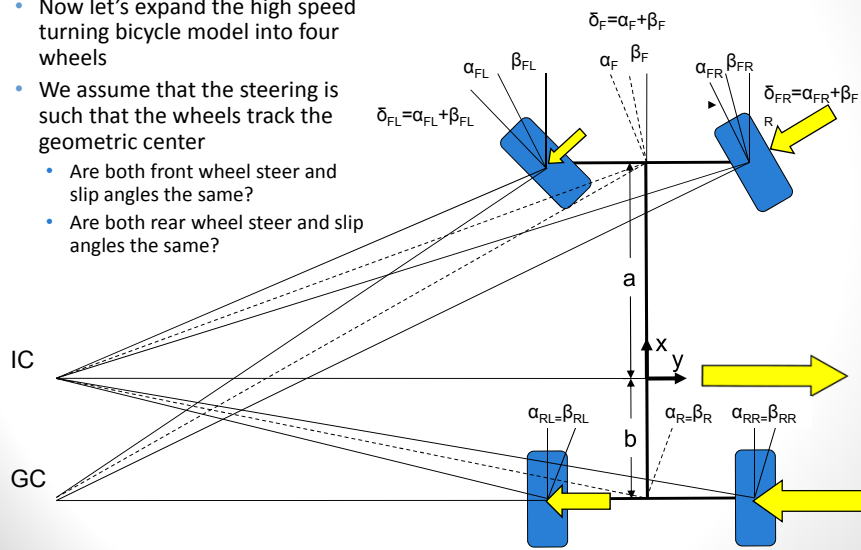


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Simple Four Wheel Model

- Now let's expand the high speed turning bicycle model into four wheels
- We assume that the steering is such that the wheels track the geometric center
 - Are both front wheel steer and slip angles the same?
 - Are both rear wheel steer and slip angles the same?



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Steady State Vehicle Dynamics

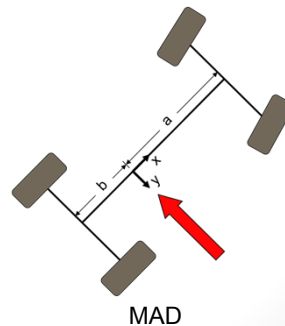
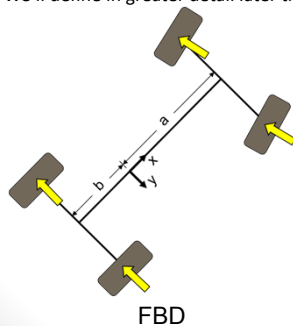
- Remember the force balance for steady state cornering

$$F_{yFL} + F_{yFR} + F_{yRL} + F_{yRR} = m * latG$$

and if the car is neutral

$$(F_{yFL} + F_{yFR}) * a = (F_{yRL} + F_{yRR}) * b$$

- What does neutral mean?
 - The lateral acceleration on the front and rear are equal
 - What do we call the non-neutral conditions?
 - Understeer
 - Oversteer
 - We'll define in greater detail later these later...



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Introduction to Pneumatic Tires

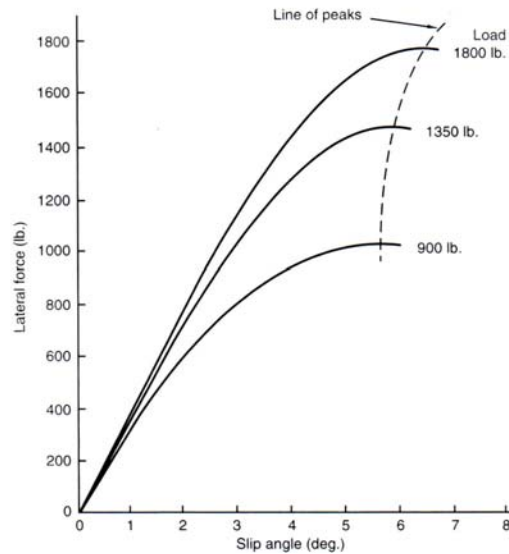
- Tires generate lateral forces or lateral grip as a result of the lateral deformation resulting from the tire rolling in a direction different from the heading of the vehicle
- How do tires behave under different normal loads and slip angles?

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Lateral Force vs. Slip Angle

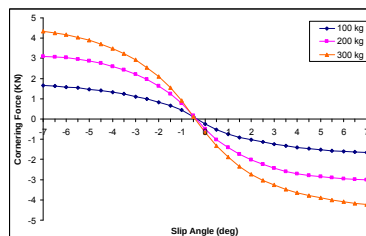
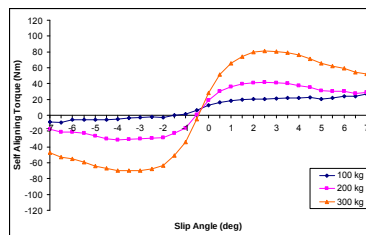
P215/60 R15 Goodyear Eagle GT-S (shaved for racing) 31 psi.



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Tire Feedback



- So how do you know you are at the maximum adhesion?
 - Is it when the torque required through the steering wheel falls off?
- Consider the data to the left
 - Upper: Aligning Moment
 - Lower: Lateral Force
 - The maximum grip is at a greater slip angle than the maximum self-aligning torque
 - Is this a good thing?
 - What does this tell you about
 - Your average driver feedback
 - Your race driver's feedback
- What do you notice about the relationship of normal load and lateral grip? Slip angle?

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Some Tire Related Questions

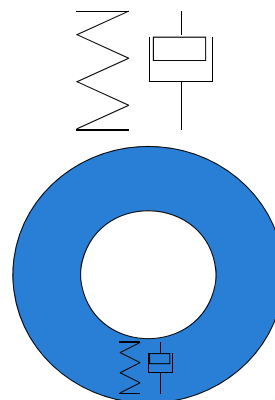
- What is the relationship you want between the steer angles of the front wheels?
 - Same?
 - Greater inside?
 - Greater outside?
- The rear wheels?
- Do you see that to manipulate the way the car responds we need to have a fundamental understanding of how pneumatic tires work?

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What about road variations?

- Pneumatic tires are
 - Springs
 - Dampers
- We also incorporate springs and dampers into most vehicles
- What is the requirement that leads us to include suspension?



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What about road variations?

- For passenger vehicles we want to isolate the occupants (or payload) from road irregularities
 - Minimize the accelerations of the passenger/cargo due to disturbances
- What about racing vehicles?
 - Want to maximize grip
 - Under what conditions do you think grip is maximized?
 - Under varying load a tire will lose as much as 60% of it's grip!
 - Therefore, minimize the load variation at the tire/ground plane for maximum grip

